

Machine Learning II - Structured Models

Exercise sheet **Duality in Convex Programming. Linear Programs**

Bogdan Savchynskyy - Bogdan.Savchynskyy@tu-dresden.de

Exercise 1 (Lagrange Dual). Construct the Lagrange dual for the problem:

$$\min -x_2 \tag{1}$$

$$x_1 - x_2 - 1 \geq 0 \tag{2}$$

$$x_1 + x_2 - 1 \geq 0 \tag{3}$$

$$x_1 \leq 2 \tag{4}$$

Compute the optima of the primal and dual problems. Are they equal to each other?

Exercise 2 (Lagrange Dual of LP). Construct the Lagrangian dual for the linear problems and check that they constitute LP as well.

1.

$$\min_x x_1 + 2x_3 - x_4 \tag{5}$$

$$x_1 + x_2 = 1 \tag{6}$$

$$x_2 + x_3 + x_4 = 0 \tag{7}$$

$$x_1 \geq 0 \tag{8}$$

$$x_4 \geq 0 \tag{9}$$

Select any feasible point of the primal and dual problems and evaluate their objectives in these points. Check that the primal objective value is not smaller than the dual one.

2.

$$\min_x \langle \theta, x \rangle \tag{10}$$

$$Ax = b \tag{11}$$

3.

$$\min_x \langle \theta, x \rangle \tag{12}$$

$$Ax = b \tag{13}$$

$$x_1 \geq 0 \tag{14}$$

4.

$$\min_x \langle \theta, x \rangle \tag{15}$$

$$Ax \geq b \tag{16}$$

$$x \geq 0 \tag{17}$$

Fill out the Table with your results.